

Terrain-Conditioned Diffusion for Generating Street-Level Expansion Layouts of Small Towns

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Code: <https://github.com/eshankiyer/terrain-urban-diffusion>

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Abstract

Generative models for urban layouts have concentrated on large, grid-dominated cities, typically in the United States, and almost universally ignore the terrain the city sits on. Yet for the small towns that make up most settlements worldwide—and especially for towns in hilly regions of Europe and Asia—topography is a first-order constraint on where streets and buildings can go. We present a conditional denoising diffusion model that generates street-level expansion layouts (road rasters and continuous built-up density) for small towns, conditioned jointly on terrain (elevation and slope) and the observed settlement state. The model is trained on windows around approximately sixty towns in hilly terrain across Europe and Asia, assembled automatically from OpenStreetMap vectors, open elevation tiles, and the multi-epoch GHS-BUILT-S built-up surface, whose 1980–2020 epochs provide real temporal supervision: each sample conditions on the town as observed at one epoch and targets the next. A concentric-erosion proxy that manufactures pseudo-temporal pairs from a single snapshot, used in an earlier version of this work, is retained as a baseline. An eleven-metric raster scorecard—covering 15-minute amenity access, infill, land efficiency, hazard avoidance, green preservation, and access equity—turns the sampler into an optimizer by best-of- N selection, and post-hoc classifiers annotate selected plans with bike infrastructure and typed zones. Training converges from 0.744 to approximately 0.04 noise-prediction loss over 120 epochs; we show top-ranked generated futures for a held-out town with per-pixel walk-time maps, specify the held-out evaluation protocol, and release code, the town list, and the full pipeline. A single trained model supports proposal generation for real towns, iterative sandbox growth on arbitrary heightmaps, and continuation of user-drawn plans.

1 Introduction

Most generative work on urban form asks the question: what does a city look like? This paper asks a narrower and, we argue, more practically useful question: given a particular small town, on particular terrain, what would its next increment of growth plausibly look like?

The distinction matters for three reasons. First, most of the world’s settlements are not large cities. Small towns dominate by count, and their growth decisions—where a new residential street goes, which hillside gets terraced—are made with far less planning capacity than a metropolitan government has. A tool that proposes terrain-respecting expansion options is more useful at this scale, not less. Second, terrain is the binding constraint precisely where planning capacity is lowest: the hill towns of Nepal, the Italian Apennines, or the Western Ghats cannot borrow street patterns from Chicago. Third, existing generative models have a documented geographic bias: the most capable urban layout generators are trained on large, mostly American cities [2, 3, 4], whose

gridded morphology transfers poorly to organically grown towns; where non-US cities have been included, the models have measurably struggled with non-grid street orientation [5].

Terrain-aware settlement growth modelling is not itself new. The SLEUTH family of cellular automata has used slope as a growth driver since the 1990s [1], and modern deep-learning land-use-change models routinely include elevation among their covariates [6]. But these models predict growth *probability* at coarse resolution: their output is a per-pixel likelihood of urbanisation, not a design. They cannot say where the street goes. Conversely, recent diffusion-based layout generators produce street-level designs but condition on land use, text, or nothing at all — not on topography [2, 7, 8]. Diffusion models that do engage with terrain generate the terrain itself [9, 10], not settlements upon it. The specific combination this paper addresses—street-level generative layout design, conditioned on terrain, for small towns, trained on European and Asian data—is, to our knowledge, unoccupied.

Our contributions are:

1. A fully automatic data pipeline that assembles terrain-conditioned growth training pairs for arbitrary towns from OpenStreetMap, open elevation tiles, and the multi-epoch GHS-BUILT-S built-up surface [21], requiring no manual annotation and no API keys, together with a curated list of ~ 60 training and 8 held-out small towns in hilly terrain across Europe and Asia. Supervision is real temporal change between observed epochs (1980–2020); the concentric-erosion proxy of the first version of this work, which recovers pseudo-temporal pairs from a single snapshot, is kept as a baseline and ablation, and we state the temporal proxy that remains (roads) explicitly.
2. A compact ($\sim 13\text{M}$ parameter) conditional diffusion model that generates joint road/built-density growth rasters, where density is continuous so that infill densification and greenfield sprawl are both supervised and both generable.
3. An eleven-metric sustainability scorecard, computed directly on rasters and the extracted road graph against real OSM amenities, greenspace, and terrain hazard proxies, that turns the sampler into an optimizer via best-of- N selection without retraining.
4. Post-hoc annotation stages—a slope-aware bike-infrastructure edge classifier and a typed-zone classifier—that add planning detail the diffusion targets cannot carry, each with a leave-one-town-out transfer protocol.
5. A demonstration that one trained model supports proposal generation for real towns, iterative game-like growth of synthetic settlements on arbitrary heightmaps, and (in the v3 extension) continuation of user-drawn plans, which we discuss as the decision-making layer of a prospective automated city-building agent.

2 Related Work

2.1 Generative urban layout models

Procedural methods based on shape grammars and production rules have long been used to synthesise street networks [11, 12], and remain the standard in game tooling; their weakness is the manual rule-crafting required per style. Learning-based approaches replaced rules with data: Neural Turtle Graphics models road layouts autoregressively [13], GlobalMapper generates building layouts for arbitrary blocks with graph attention [4], and CityGen [3] and CityDreamer [14] produce unbounded city layouts with diffusion. Closest to our setting, He et al. propose a stepwise

multimodal-diffusion urban design framework trained on Chicago and New York [2], and explicitly note the restriction to US-grid morphology as a limitation. GAN-based urban form models trained on multiple cities report degraded fidelity on non-grid cities [5]. None of these condition on terrain.

2.2 Urban growth and land-use-change models

SLEUTH [1] takes slope, land use, exclusions, urban extent, transport and hillshade as inputs to a cellular automaton and remains widely applied [15]; extensions couple CA with support vector machines [16] or deep networks [6]. Deep sequence models (ConvLSTM) forecast urban growth from satellite time series [17]. These models incorporate terrain but output coarse urbanisation probabilities rather than street-level layouts; they simulate *whether* a cell urbanises, not *what is built*. Our temporal supervision draws on the same class of multi-epoch built-up products these models consume [21], but uses them as diffusion targets rather than CA calibration data.

2.3 Terrain and geospatial generation

Diffusion models generate realistic terrain from text or sketches [9, 10, 24], and map-or-primitive-conditioned diffusion synthesises satellite imagery [7, 22]. GPS coordinates have been used as conditioning for image generation [23]. This line of work treats terrain as the *output* or the imagery as the target; we invert the relationship, treating terrain as a constraint under which settlement structure must be generated.

2.4 Generative models as game agents

Automating construction in city-building games has mostly meant procedural map generation. Playing such a game requires perception, decision-making, and actuation layers. We position our model as the decision-making layer—current city state and heightmap in, next construction increment out—and discuss integration via game modding APIs in Section 8.

3 Method

3.1 Problem formulation

Let $E \in \mathbb{R}^{H \times W}$ be an elevation raster over a window of ~ 1.9 km ($H=W=128$ at 15 m/px), S its slope, $D_t \in [0, 1]^{H \times W}$ the observed built-up density at epoch t (the built fraction of each cell), and R_t a binary raster of the road network inside the epoch- t settlement footprint. The model learns the conditional distribution $p(R_{\text{new}}, D_{t+1} \mid E, S, D_t, R_t)$ over roads outside the epoch- t footprint and the next-epoch density field. Conditioning channels are concatenated: elevation is z-scored per window and clipped to $\pm 3\sigma$; slope is normalised by 30° ; density is used directly. Targets are affinely mapped to $[-1, 1]$ (binary road targets to $\{-1, 1\}$). Because density is continuous rather than a binary footprint, the target expresses both infill densification (cells that were already built getting denser) and greenfield sprawl (new cells), and both are supervised.

3.2 Temporal supervision from multi-epoch built-up data

The primary training regime replaces manufactured growth pairs with observed ones. GHS-BUILT-S R2023A [21] provides a global built-up surface at 3 arc-seconds on the WGS84 tile grid for the epochs 1980, 1990, 2000, 2010, and 2020; our pipeline downloads and caches the relevant 10° tiles once and performs windowed reads per town. For each town and each consecutive epoch pair

$(t, t+1)$ with sufficient observed growth we emit one sample: the model conditions on elevation, slope, the observed density D_t , and roads clipped to the epoch- t footprint, and targets D_{t+1} together with roads outside that footprint. Windows whose epoch pair shows too little change are dropped automatically.

One honest proxy remains. OpenStreetMap carries no road history for these towns, so “roads at epoch t ” are necessarily present-day OSM roads restricted to where density was already non-zero at epoch t . Roads built later inside an old footprint therefore leak into early-epoch conditioning; we flag the consequences in Section 9. The built-up channels, which carry the growth signal itself, are genuinely temporal.

3.3 The concentric-erosion proxy (v1 baseline)

The first version of this work constructed supervision from present-day snapshots alone: the observed built-up mask B was eroded with a disc of radius $\rho = 10$ px (≈ 150 m) to produce a pseudo-historical core B_c , and the ring $B_r = B \setminus B_c$ became the generation target, with ring roads defined analogously. The proxy assumes predominantly outward, concentric growth—reasonable for compact hill towns whose steep surroundings suppress leapfrog development, wrong for linear valley towns and for growth dominated by infill, and structurally incapable of expressing densification, since its targets are binary. We retain it as a baseline and ablation: the channel counts match the temporal regime, so the same architecture trains on either supervision unchanged, which is precisely what makes the comparison clean.

3.4 Architecture and training

A U-Net with three resolution levels (64/128/256 channels), residual blocks, sinusoidal timestep embeddings, and self-attention at the two coarsest resolutions predicts the added noise ϵ under a cosine noise schedule with $T=1000$ steps [18, 19]. The 4 conditioning channels are concatenated to the noisy target channels at the input (~ 13 M parameters). We train with AdamW, cosine learning-rate decay, gradient clipping at 1.0, and an EMA of the weights (decay 0.999); sampling uses 50 DDIM steps [20]. Each town window is augmented by the dihedral group (8 orientations) and small centre jitter.

3.5 Inference modes

Real mode. For a real town, the pipeline fetches current elevation, OSM rasters, and the latest GHSL epoch, and samples a next-epoch density and road increment conditioned on the true terrain and observed state. **Sandbox mode.** For an arbitrary heightmap (fractal noise here; a game-exported heightmap in principle), a seed settlement of a few pixels is placed on low-slope ground and the model is applied iteratively: the union of the previous state and the generated increment becomes the next conditioning state. Each round is one “turn” of automated play. The plan-import mode of Section 6 generalises this by letting a human drawing, rather than a seed, initialise the state.

4 Sustainability evaluation and best-of- N selection

A generative model of plausible growth is not yet a planning tool: a planner needs to know which of many plausible futures is *better*. We therefore attach an eleven-metric raster scorecard that grades any (roads, density) plan against its terrain and environmental context, and use it for best-of- N

selection: sample N futures from the diffusion model, score each, keep the top-ranked ones. This turns the sampler into an optimizer without retraining, and the same scorecard grades real draft plans for critique.

The context layers are fetched once per town: real OSM amenities in seven categories (food, education, health, civic, recreation, transit, social), greenspace and water masks, and two terrain hazard proxies derived from the DEM. The metrics, each normalised to $[0, 1]$, are:

- **15-minute amenity coverage** (weight 0.18): per-pixel network walk times are computed by Dijkstra over road pixels at 80 m/min with a 15-minute (1200 m) budget, with real OSM amenities as fixed destinations; coverage is the built-weighted served fraction over the amenity categories *present in the window*, so small towns are compared on what exists. A category that exists but is unreachable over the network contributes zero while remaining in the denominator.
- **Infill share** (0.10): fraction of new growth inside the existing footprint rather than greenfield.
- **Land efficiency** (0.08): added density per newly consumed pixel.
- **Circuitry** (0.06): network over straight-line distance between sampled road-pixel pairs.
- **Earthwork** (0.06): mean terrain slope under *new* roads, a grading-cost and erosion-risk proxy.
- **Green preservation** (0.10): survival of *baseline* functional green patches (≥ 0.45 ha); new green elsewhere cannot buy back cleared green, and new roads clear green just as buildings do.
- **Green access** (0.08): share of built pixels within a 300 m network walk of a remaining functional green patch.
- **Flood avoidance** (0.10): new development on flood-prone cells, flagged by a HAND proxy—height above the nearest mapped water under 5 m within ~ 105 m of water.
- **Landslide avoidance** (0.08): new development on slopes above 25° , with slopes above 35° counted double.
- **Congestion proxy** (0.08): one minus the Gini coefficient of edge betweenness on the road graph extracted from the raster, so plans that funnel all traffic through one link score low.
- **Access equity** (0.08): population-weighted dispersion of per-cell amenity coverage, penalising plans that serve the centre and strand the periphery.

Weights sum to one. Hazard and earthwork terms are computed over new development only, so a town’s pre-existing switchbacks are not held against a plan. Ranking applies a growth-delivery factor, so a do-nothing plan cannot win best-of- N simply by avoiding every hazard: the scorecard rewards delivering growth well, not refusing to grow.

Two boundaries are deliberate. The metrics are spatial-form proxies computed from rasters—HAND and slope thresholds are not hydraulic or geotechnical models, and no score here certifies environmental performance. And socioeconomic factors—income, tenure, affordability, displacement risk—are named as required human inputs and deliberately not fabricated from rasters. As a methods-transparency note: the metric design came from a multi-perspective specification (ecologist, transport engineer, geotechnical engineer, equity planner) and the implementation survived

a two-round adversarial verification loop that caught one critical defect—pre-existing roads scored as new development—before acceptance.

5 Post-hoc annotation stages

Two planning layers are kept out of the diffusion targets because their OSM labels are too sparse in the training towns to supervise generation; both are instead trained where labels are dense and applied to generated rasters, with leave-one-town-out protocols so the transfer claim is testable rather than asserted.

5.1 Bike-infrastructure edge classification

Cycleway tagging is dense in the flat cycling countries and nearly absent in mountain towns, so bike lanes are a per-edge classifier, not a generated channel. We train a logistic classifier on twelve well-tagged flat European towns (Dutch, Danish, and north-German), using physical features that transfer across morphology: mean absolute grade along the edge computed from the DEM, road class, edge length, and the edge’s position relative to the town centre. Evaluation is leave-one-town-out. At inference, a skeleton-to-graph extractor converts any generated road raster into junction-to-junction polylines, and every edge receives a bike-infrastructure probability—so a plan selected in Section 4 arrives annotated with the edges where protected cycling infrastructure is physically plausible.

5.2 Typed zones

The diffusion model generates one undifferentiated density field; a planner needs to know where residential, commercial, industrial, and institutional uses go. OSM landuse polygons provide labels, but they are patchy outside Europe and heavily imbalanced toward residential, so zoning is likewise a post-hoc per-pixel classifier trained only where labels exist, on features the generator already produces (distance to centre, slope, road proximity, local density, amenity density, local road density), with per-class caps against residential dominance. Evaluation is leave-one-town-out macro-F1; at inference the classifier paints use classes onto new development.

6 Extensions in progress (v3)

Three extensions are implemented in the released pipeline and currently training; we describe them as method and claim no results for them.

Generated amenity channel. The v2 scorecard measures access against *existing* amenities only, which correctly penalises remote growth but cannot express “grow here and put a school there.” v3 adds a third target channel of log-scaled amenity density rasterised from OSM points, so the model learns where local centres sit relative to fabric and, at sampling time, *proposes* amenity locations for new growth. The scorecard reports access both with and without the proposed amenities, so the assumption stays visible rather than laundering optimism into the headline score.

Plan import. A user-drawn plan image following a simple convention—black lines for roads, warm colours for built area, green kept free—is parsed into conditioning rasters and continued by the model, over real terrain, a supplied heightmap, or procedural relief. This generalises sandbox mode: the initial state comes from a human drawing instead of a seed pixel, and the conditioning layout is unchanged, so any v2/v3 checkpoint works.

Plan-style rendering. Raw model rasters are hard for humans to read, so a rendering stage composes hillshaded terrain, roads, zone colours, green and water, and proposed local centres into a planning-map figure with a legend and scale bar.

7 Experiments

7.1 Data

~60 training towns and 8 held-out towns (population roughly 5k–60k) in hilly regions of Europe (Alps, Apennines, Iberia, Balkans) and Asia (Japan, the Indian Himalaya and Western Ghats, Nepal, Vietnam, Indonesia, Turkey, the Caucasus), listed in the repository. Each town contributes up to four consecutive GHSL epoch pairs (1980→1990 through 2010→2020); pairs with insufficient observed growth are dropped automatically, and each surviving window is expanded eightfold by dihedral augmentation. The exact per-run sample manifest is produced by the released pipeline.

7.2 Evaluation protocol

On each held-out town the protocol compares the generated increment to the real next-epoch change: **Slope KS** — Kolmogorov–Smirnov distance between the slope distributions under generated vs. real new built-up pixels (lower is better; measures whether the model builds on the kinds of slopes the town actually built on); **Road connectivity** — fraction of generated road pixels 8-connected to the existing network; **Built contiguity** — fraction of new built pixels adjacent to at least two settlement pixels; **Volume ratio** — generated/real new built area; plus ordinal density-class agreement derived by binning the continuous density channel. Two comparisons are planned: a slope-blind baseline that scatters the same built mass uniformly at random over the window (a deliberately weak control that isolates whether the model uses terrain at all), and the erosion-trained v1 model as an ablation of the temporal supervision. The protocol is implemented in the released pipeline; we do not report held-out numbers in this version because that evaluation run has not completed, and we prefer an empty cell to an invented one. For the post-hoc stages, the corresponding protocols are leave-one-town-out AUC (bike edges) and leave-one-town-out macro-F1 (zones), likewise shipped with the code.

7.3 Results

Training. The v2 temporal regime trains stably: the noise-prediction MSE falls from 0.744 at the first epoch to ~0.040 by epoch 120 (final-epoch 0.036, mean of the last ten epochs 0.039), with no divergence and the familiar fast-then-slow diffusion profile (Figure 1). We read this only as evidence that the continuous-density temporal targets are learnable by the unchanged v1 architecture; training loss is not a fidelity claim.

Qualitative selection results. Figure 2 shows the pipeline end to end on a held-out town: many futures are sampled, the scorecard of Section 4 ranks them, and the top three are shown with their per-pixel walk-time maps and bike-lane annotations. The three selected futures score within one point of each other (51, 50, 50 of 100) while proposing visibly different spatial arrangements, and their 15-minute amenity coverage of built pixels ranges from 82% to 87%—which is the intended behaviour of best-of- N selection: several near-equivalent good options for a human to choose among, not a single oracle answer. The density fields are speckled at the pixel level, as raw diffusion rasters are; the rendering stage of Section 6 exists precisely because these rasters are for scoring, not for reading.

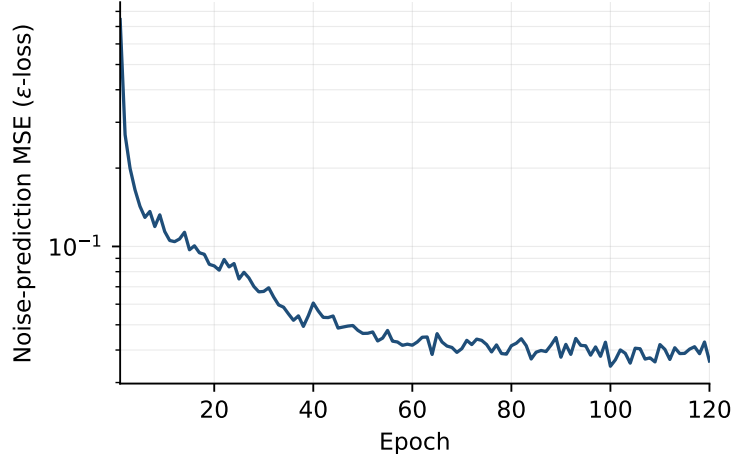


Figure 1: Training loss of the v2 model (noise-prediction MSE, log scale) over 120 epochs on the GHSL temporal pairs: 0.744 at epoch 1 to ~ 0.040 at epoch 120.

8 Applications: towards automated city-building agents

The sandbox mode is a decision-making layer for game automation: at each turn it maps (heightmap, current city) to a construction increment. A complete agent for a title such as *Cities: Skylines 2* additionally needs perception (reading the game’s heightmap and current road/zone state, available through its modding API) and actuation (executing placements, also exposed to mods). Because our model consumes and produces rasters, the integration surface is thin, and the plan-import mode of Section 6 already parses drawn or exported maps into conditioning. We do not implement the agent here; we note that raster-to-game-geometry conversion (vectorising road rasters into placeable splines) is the main engineering step, and the skeleton-to-graph extractor of Section 5 is most of it.

9 Limitations

Temporal road proxy. OSM has no road history, so epoch- t roads are present-day roads clipped to the epoch- t footprint (Section 3.2). Roads built recently inside an old footprint leak into early-epoch conditioning, making the conditioning network denser than the town actually was; the density channels are unaffected. **Density granularity.** GHS-BUILT-S is native ~ 100 m (3 arc-second) resolution resampled to our 15 m/px grid, so the density target is smoother than the road raster and cannot carry parcel-level structure; the road channel and the 15 m DEM carry the fine detail. **Label sparsity outside Europe.** OSM amenity, landuse, and cycleway coverage thins outside Europe, so the scorecard’s destination set is less complete in the Asian towns and the post-hoc classifiers are trained on European labels; leave-one-town-out transfer is tested only within that labelled pool. **Window scale.** 1.9 km windows exclude transit-scale structure—tram or rail corridors, and anything that only makes sense at city scale—which is why transit appears in the scorecard only as a destination category, not as generated infrastructure. **Scorecard scope.** The eleven metrics are spatial-form proxies, not certified environmental performance (Section 4); no zoning law, land ownership, budget, or hydraulic model enters the pipeline, and generated layouts are proposals, not plans. **Baselines.** The slope-blind control is deliberately weak; comparison against SLEUTH-class

CA models and layout generators retrofitted with terrain channels is future work.

10 Conclusion

We presented a terrain-conditioned diffusion model for street-level growth of small towns, trained on real 1980–2020 built-up change assembled fully automatically from open data, scored and selected by an eleven-metric sustainability scorecard, annotated post hoc with bike infrastructure and typed zones, and usable as a proposal tool for real settlements, an iterative growth engine on arbitrary heightmaps, and a continuation engine for human-drawn plans.

AI assistance disclosure

The code, experiments, and text of this paper were produced with substantial assistance from an AI system (Anthropic Claude), directed and reviewed by the author. This disclosure is made in accordance with venue policies on AI-assisted research.

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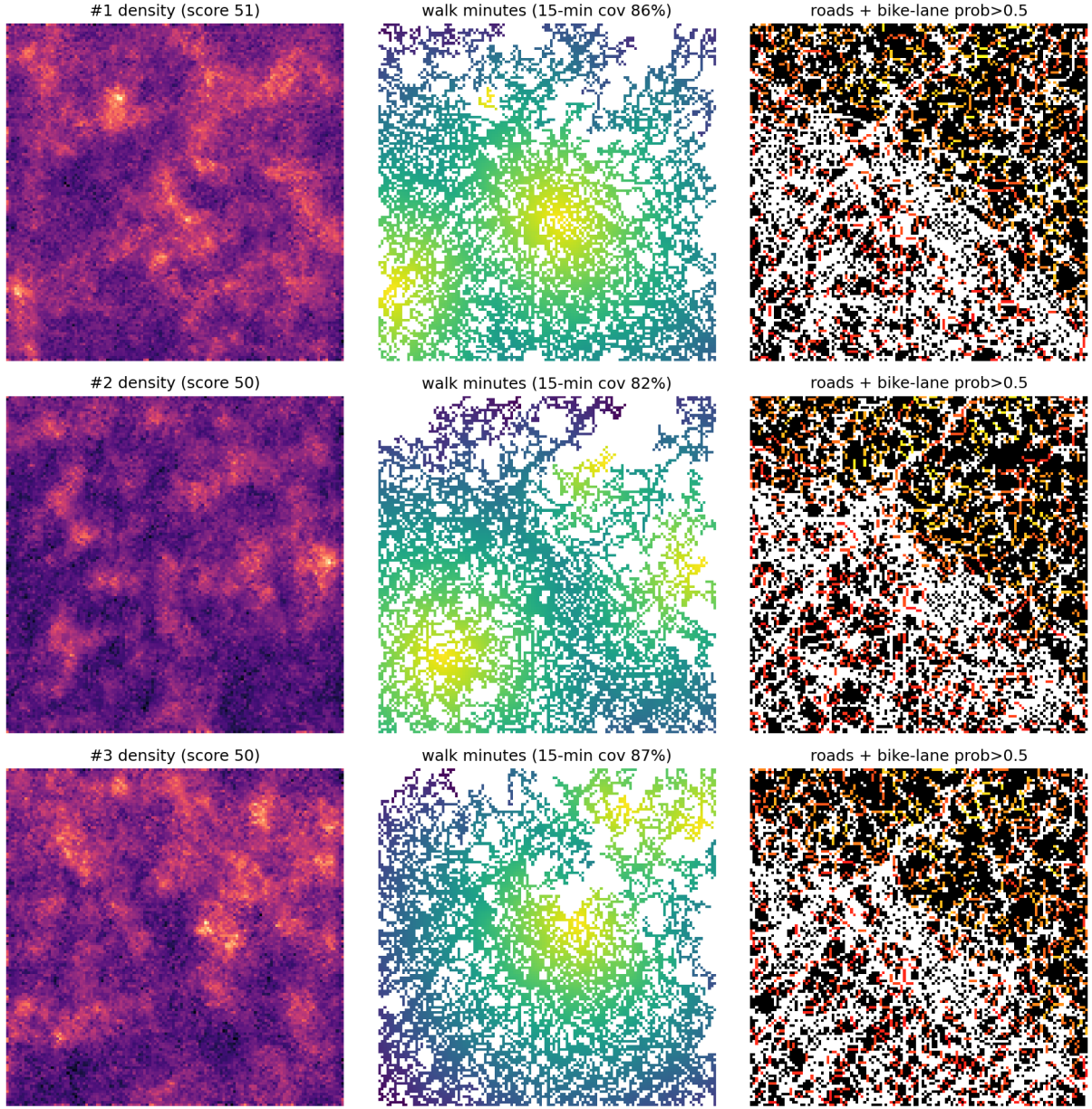


Figure 2: Top-3 of N sampled futures for a held-out town, ranked by the eleven-metric scorecard (rows; total scores 51, 50, 50 of 100). Left: generated continuous built-density channel. Centre: per-pixel walk time along the generated road network to the nearest real OSM amenity (Dijkstra at 80 m/min), with the share of built pixels inside the 15-minute budget given per sample (86%, 82%, 87%). Right: road pixels (black) with graph edges whose predicted bike-infrastructure probability exceeds 0.5 overlaid in warm colours by the post-hoc classifier of Section 5. These are raw model rasters, deliberately shown without the plan-style rendering stage.